

Arjuna JEE 2.0 2027

Maths

Basic Mathematics

DPP: 2

Total Duration - 25 Min

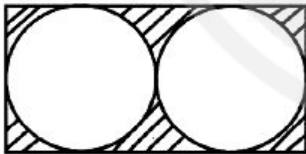
Q1 "Zero" is

- (A) positive number
- (B) negative number
- (C) both positive as well as negative
- (D) neither positive nor negative

Q2 If $\sqrt{x^2 - 6x + 9} + \sqrt{e^y - 1} + \sqrt{\tan^2 \theta - 3} = 0$
then the value of $x + y + \sec^2 \theta$ is

- Q3 If $x + \frac{1}{x} = a$, $x^2 + \frac{1}{x^3} = b$, then $x^3 + \frac{1}{x^2}$ is
- (A) $a^3 + a^2 - 3a - 2 - b$
 - (B) $a^3 - a^2 - 3a + 4 - b$
 - (C) $a^3 - a^2 + 3a - 6 - b$
 - (D) $a^3 + a^2 + 3a - 16 - b$

Q4 The ratio of total area of the rectangle to the total shaded area



- (A) $\frac{2}{\pi}$
 - (B) $\frac{4}{4-\pi}$
 - (C) $\frac{4-\pi}{\pi}$
 - (D) $\frac{\pi}{4}$
- Q5 What is the area of an equilateral triangle inscribed in a circle of radius 4 cm?

- (A) 12 cm^2
- (B) $9\sqrt{3} \text{ cm}^2$
- (C) $8\sqrt{3} \text{ cm}^2$
- (D) $12\sqrt{3} \text{ cm}^2$

Q6 If $x = 3 + 3^{1/3} + 3^{2/3}$, then the value of $x^3 - 9x^2 + 18x - 12$ is

- (A) 0
- (B) -1
- (C) 1
- (D) 2

Q7 If $x + \frac{1}{x} = 2$, then prove that

$$x^2 + \frac{1}{x^2} = x^4 + \frac{1}{x^4} = x^6 + \frac{1}{x^6}$$

Q8 If $2x - 5y = 13$ and $xy = 5$, then the value of $8x^3 - 125y^3$ is

- (A) 4211
- (B) 4130
- (C) 4126
- (D) 4147

Q9 The value of

$$\frac{(1.5)^3 + (4.7)^3 + (3.8)^3 - 3 \times 1.5 \times 4.7 \times 3.8}{(1.5)^2 + (4.7)^2 + (3.8)^2 - 1.5 \times 4.7 - 1.5 \times 3.8 - 4.7 \times 3.8}$$
 is

- (A) 8
- (B) 9
- (C) 10
- (D) 11

Q10 If $x^2 + \frac{1}{x^2} = 51$, then what is the value of $x^3 - \frac{1}{x^3}$

- (A) 364
- (B) 365
- (C) 756
- (D) 367



Answer Key

Q1 (D)

Q2 7

Q3 (A)

Q4 (B)

Q5 (D)

Q6 (A)

Q7 Proof

Q8 (D)

Q9 (C)

Q10 (A)



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